

HSN Code 8482 – Import & Export Forecasting (2014–2024)

1. Introduction

This project develops a predictive model for trade flows (Imports and Exports) of HSN Code 8482 (bearings) for India. The aim is to forecast future trade values using time series models (ARIMA & ARIMAX).

2. Dataset

Period: 2014–2024

Source: Trade data for HSN 8482

Variables used:

Trade Flow (Imports = M, Exports = X)

Trade Value (US\$)

3. Methodology

Data Cleaning:

Converted trade values to numeric format

Mapped trade flows: M → Import, X → Export

Created pivot table by year

Modeling Approach:

Imports: ARIMA(1,1,1)

Exports: ARIMAX(1,1,1), using Imports as an exogenous variable for improved accuracy check

4. Model Performance (Accuracy Check)

Imports (ARIMA):

MAE: 70,663,385.29

RMSE: 79,375,737.08

2023 → Predicted: 1,295,058,786.35 | Actual: 1,329,567,039.00 | Error: 2.60%

2024 → Predicted: 1,293,269,077.07 | Actual: 1,400,087,595.00 | Error: 7.63%

Exports (ARIMAX):

MAE: 73,209,916.56

RMSE: 88,569,896.68

2023 → Predicted: 807,086,820.05 | Actual: 783,726,023.00 | Error: 2.98%

2024 → Predicted: 877,057,932.07 | Actual: 753,998,896.00 | Error: 16.32%

5. Forecasts (2025–2026)

Imports:

2025 → 1,293,053,838.49 US\$

2026 → 1,293,027,952.92 US\$

Exports:

2025 → 787,331,237.00 US\$

2026 → 819,132,972.86 US\$

6. Conclusion

The predictive model successfully forecasts the trade values of HSN 8482. Imports are projected to remain stable around 1.29B US\$, while exports show a growth trend from 787M US\$ in 2025 to 819M US\$ in 2026.

The use of ARIMA for imports and ARIMAX for exports provided reasonable accuracy, with error margins ranging from 2–16%. This model can be a useful tool for trade analysis and policy planning.